This receipt is from next week.

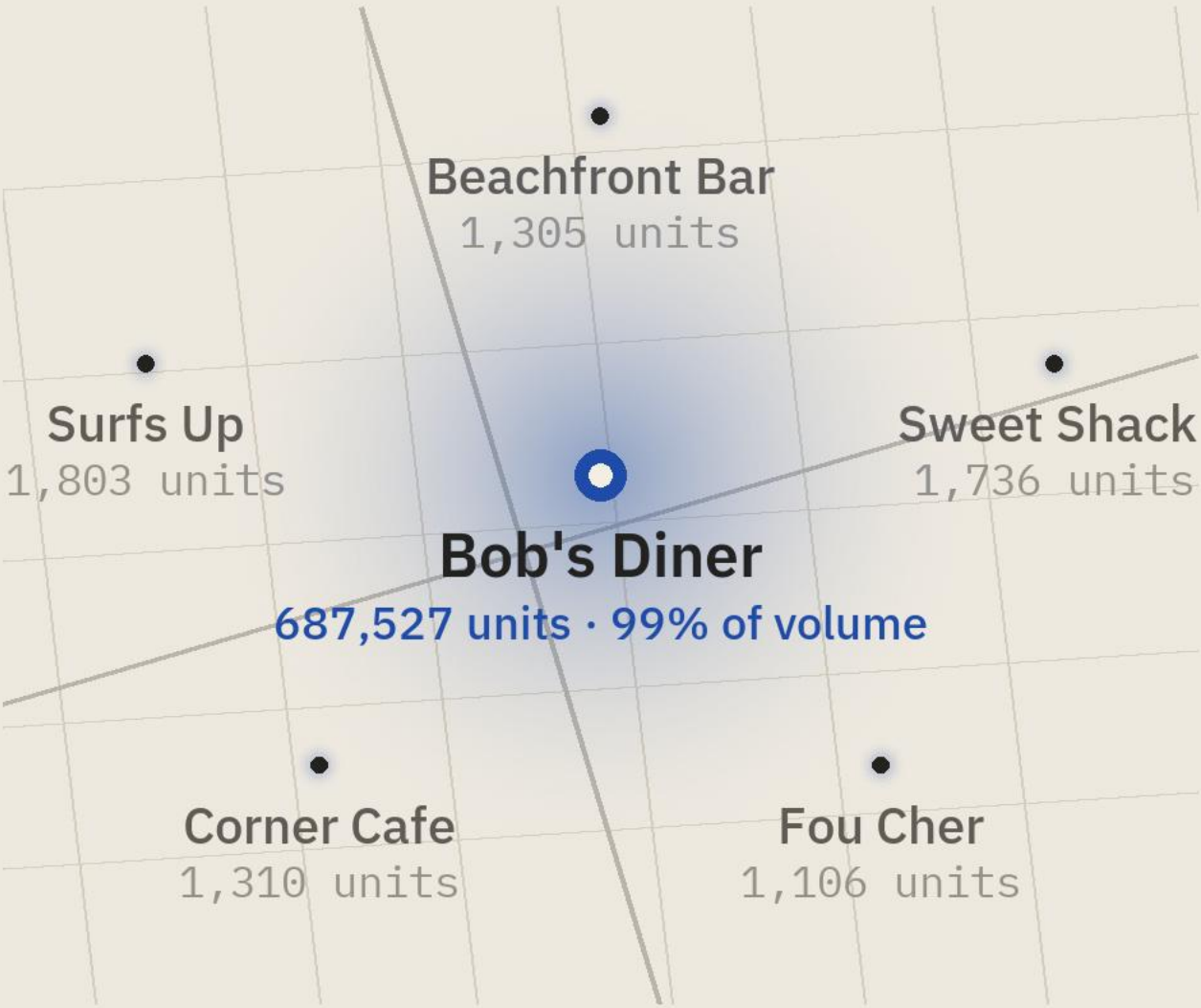
BOB'S DINER
FORECAST · NEXT 7 DAYS

Sat 2022-01-01	753.7 units
Sun 2022-01-02	292.3 units
Mon 2022-01-03	278.7 units
Tue 2022-01-04	320.1 units
Wed 2022-01-05	369.2 units
Thu 2022-01-06	548.6 units
Fri 2022-01-07	678.1 units
TOTAL	3,240.8 units

model: xgboost · MAPE 9.6%
printed 2021-12-31

Bob's Diner is 99% of volume.

Can one model fit all six?



Maximus Friedrich Baluyot

Restaurant sales forecast

Capstone 3 · Fresh Analytics

Four models, one dashboard, under 10% error

```
max@baluyot ~ $ curl -s
localhost:8104/forecast -d
'{"store":"Bob's Diner","days":7}'
```

Maximus Friedrich Baluyot



1 Sales, items, restaurants
pandas

2 Explore
matplotlib

3 Date features
pandas

4 LR · RF · XGBoost · LSTM
scikit-learn · xgboost · tensorflow

5 POST /forecast
FastAPI

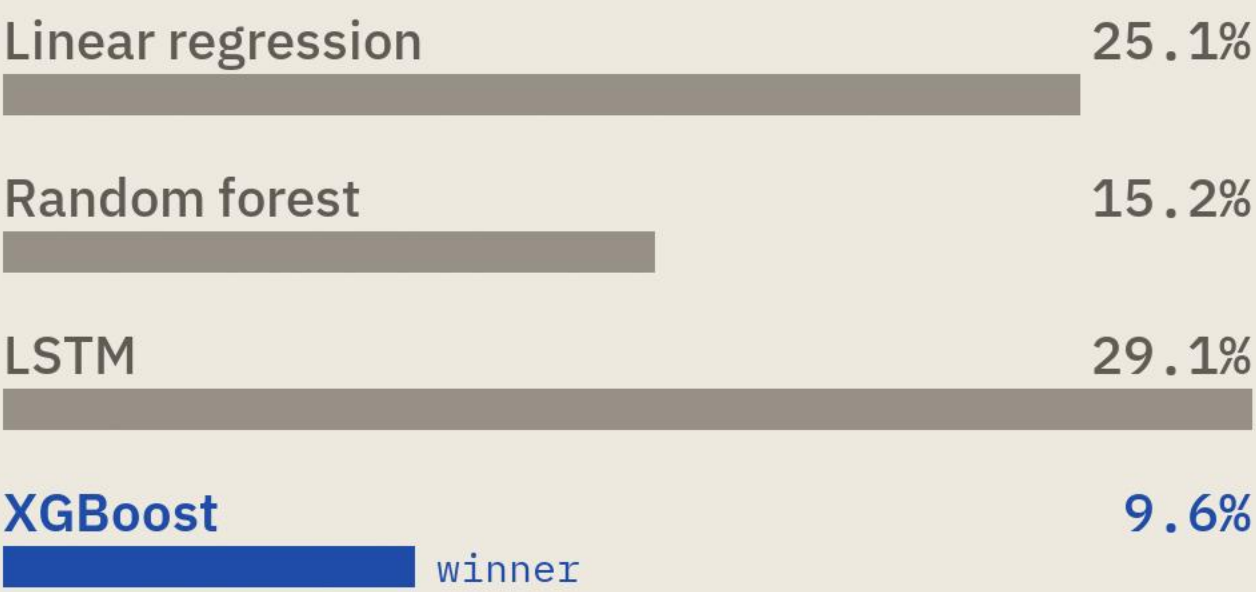
6 Dashboard
HTML + ui.css

Early stop on the last 90 training days

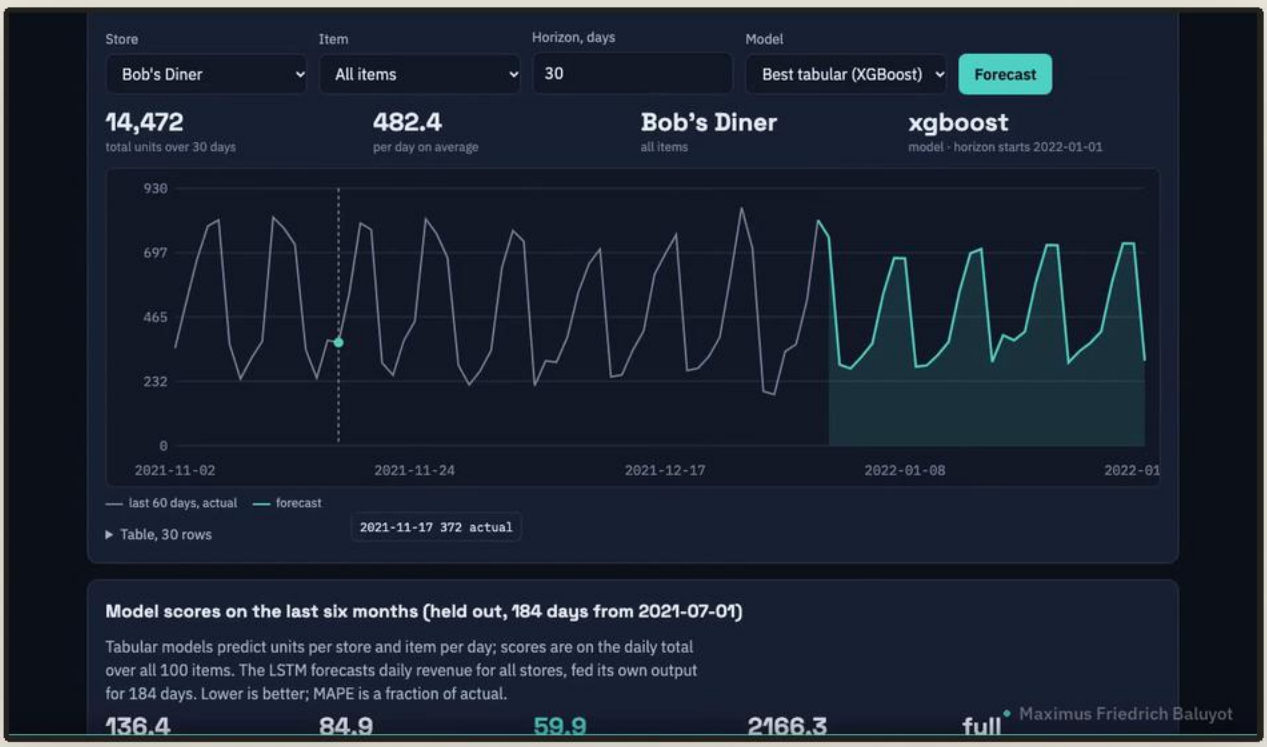
Maximus Friedrich Baluyot

Six months held out. XGBoost wins.

MAPE, daily total, 184-day holdout



Beachfront Bar, 60 days, one click



9.6%

MAPE · XGBoost · held-out six months

one limit

Carries the 2021 level forward. No trend.

Notebook 72.30 → 59.95 here. And it ships.

Maximus Friedrich Baluyot

Try it: Beachfront Bar, 60 days, Forecast.

```
max@baluyot ~ $ git clone
https://github.com/Keshigami/
capstone3-sales-forecast.git
max@baluyot ~ $ just setup
max@baluyot ~ $ just up
max@baluyot ~ $
```

github.com/Keshigami/capstone3-sales-forecast

Maximus Friedrich Baluyot

AI & ML Specialist · Applied AI · ML Engineering · Automation